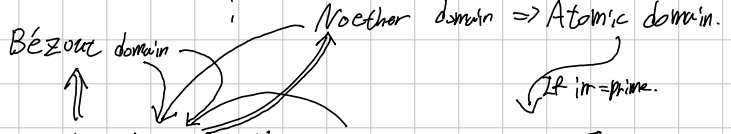
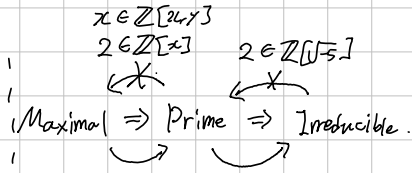
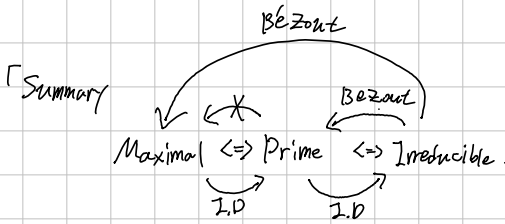
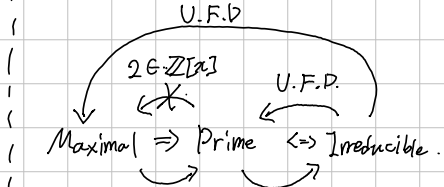
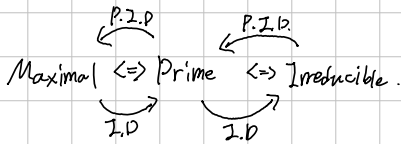


\*: Yet unproved.

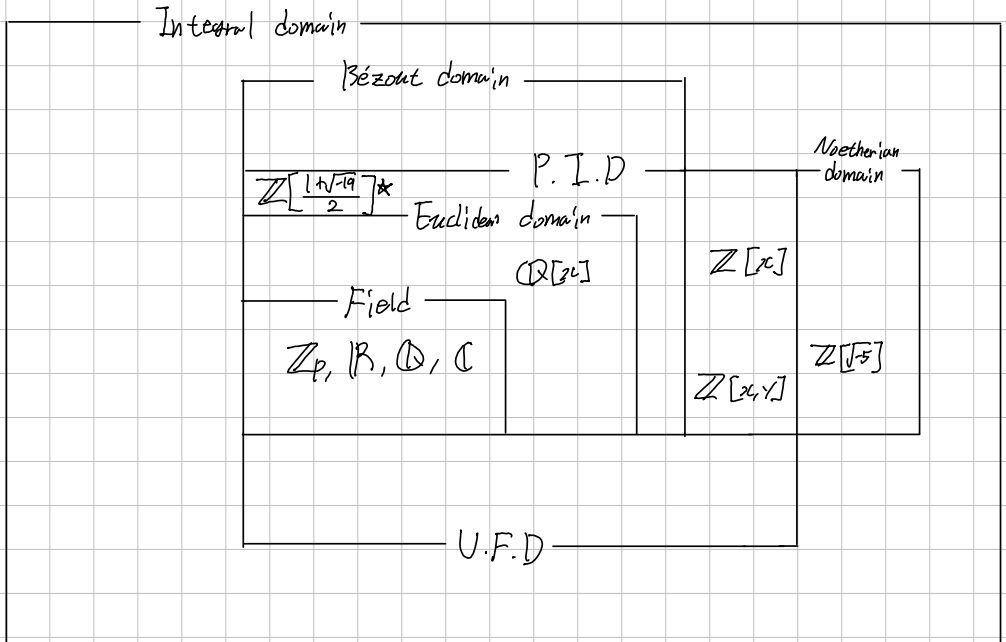


Field  $\Rightarrow$  Euclidean Domain  $\Rightarrow$  Principal Ideal Domain  $\Rightarrow$  Unique Factorization Domain  $\subset$  Integral Domain.



P.I.D.  $\Leftrightarrow$  Noether  $\Leftrightarrow$  Every ideal are finitely generated  $\Leftrightarrow$  A.C.C.I  $\Rightarrow$  A.C.C.P  $\Rightarrow$  Factorization  $\Leftrightarrow$  U.F.D.

Bézout  $\Leftrightarrow$  Finitely generated ideal is principal  $\Rightarrow$  irreducible = prime  $\Rightarrow$  ! factorization  $\Leftrightarrow$  U.F.D.



Commutative Ring with unity.

Basic Preliminaries: Let  $R$  be a Commutative Ring with unity  $1 \neq 0$ ,  $A \subseteq R$  is subset,  $I \subseteq R$  Ideal.

1).  $(A) = RA \stackrel{\text{def}}{=} \{r_1 a_1 + \dots + r_n a_n \mid n \in \mathbb{N}, r_i \in R, a_i \in A\}$

2).  $I = R$  iff  $I$  has a unit. (not necessary commutative)

3).  $R$  is field iff only Ideal of  $R$  is  $0$  or  $R$ .

4).  $P \subseteq R$  is prime  $\Leftrightarrow R/P$  is integral domain.  $\hookrightarrow \curvearrowright$

5).  $M \subseteq R$  is Maximal  $\Leftrightarrow R/M$  is field.

Def. Let  $R$  be an Integral domain, A map

$$N: R \rightarrow \mathbb{N} \cup \{0\}$$

is called a Norm if  $N(0) = 0$ . If  $N(a) > 0$  for all  $a \neq 0$ , then it is called positive Norm.

Def. Let  $R$  be a Commutative Ring, and  $a, b \in R$  with  $b \neq 0$ .

1).  $a$  is called a multiple of  $b$  if there exists  $x \in R$  s.t.  $a = bx$ . Simply denote  $b|a$ .

2). A non-zero  $d \in R$  is called a greatest common divisor of  $a$  and  $b$  if

$$\begin{cases} d|a \text{ and } d|b \\ d'|a \text{ and } d'|b, \text{ then } d'|d. \end{cases}$$

Since  $b|a \Leftrightarrow a \in (b) \Leftrightarrow (a) \subseteq (b)$ ,  $d$  is a greatest Common divisor of  $a$  and  $b$  if

$$(a, b) \subseteq (d) \quad \text{and} \quad (a, b) \subseteq (d') \Rightarrow (d) \subseteq (d'). \quad \text{Directly,}$$

Prop.  $(a, b) = (d) \Rightarrow d$  is g.c.d of  $a$  and  $b$ .

But the converse is false, In  $\mathbb{Z}[x]$ ,  $1$  is g.c.d of  $2$  and  $x$  but  $R = (1) \neq (2, x)$ .

Def. The integral domain  $R$  is said to be a Euclidean domain if: there exists a Norm  $N$  s.t. for any  $a, b \in R$  with  $b \neq 0$ , there exists  $q, r \in R$  s.t.  $a = qb + r$  where  $N(r) < N(b)$  or  $r = 0$ .

Lemma. If  $R$  is an Euclidean domain with a Norm  $N$ , then for any  $a, b \in R$  with  $b \neq 0$ , there exists a sequence  $\{r_k\}_{k=0}^{\infty}$  satisfying: put  $r_{-2} = a$ ,  $r_{-1} = b$ ,

$$\begin{cases} \text{for } 0 \leq k \leq n+1, & r_{k-2} = q_k \cdot r_{k-1} + r_k \text{ for some } q_k \in R, \\ N(b) > N(r_0) > N(r_1) > \dots > N(r_n), \\ r_{n+1} = r_{n+2} = \dots = 0. \end{cases}$$

This  $\{r_k\}_{k=0}^{\infty}$  is called remainder sequence for  $a$  and  $b$ .

Prop. Let  $R$  be an Euclidean domain. Then, for any non-zero ideal  $I \subseteq R$ ,  $I = (d)$ , where  $d$  is any non-zero element in  $I$  of minimum norm.

Proof. Let  $I \subseteq R$  be a non-zero Ideal. observe that: the set

$$N[I \setminus \{0\}] = \{N(a) \mid a \in I, a \neq 0\}$$

has a minimum element  $m \in N \setminus \{0\}$  by the Well-ordering principle,

take  $d \in N^{-1}[\{m\}]$ . Trivially,  $(d) \subseteq I$ . To show the reverse inclusion, let  $a \in I$  be given. Since  $R$  possess a Division Algorithm, for some  $q, r \in R$ ,

$$a = qd + r \quad N(r) < N(d) \text{ or } r = 0$$

But, by the Minimality of  $N(d)$ ,  $r$  must be zero. Therefore,  $a = qd \in (d)$ .  $\square$

Lemma. Let  $R$  be an Integral domain, let  $d, d' \in R$ .

If  $(d) = (d')$ , then  $d = ud'$  for some unit  $u \in R$ .

Proof. If either  $d$  or  $d'$  is zero, there is nothing to prove.

Assume both  $d$  and  $d'$  are non-zero. Since  $d \in (d')$  and  $d' \in (d)$ ,

$$d = xd' \text{ and } d' = yd \text{ for some } x, y \in R.$$

$$\text{Now, } d = xd' = xyd \text{ implies } d(1 - xy) = 0.$$

Since  $R$  has no zero divisor and  $d \neq 0$ ,  $xy = 1$ .  $x$  and  $y$  both are units.  $\square$

Thm. Let  $R$  be an Euclidean domain, and  $a$  and  $b$  be non-zero elements in  $R$ .

Let  $\{r_i\}$  be a remainder sequence for  $a$  and  $b$ . Then,

$$(r_n) = (a, b).$$

Proof. Clearly,  $r_n | r_{n-1}$  since  $r_{n-1} = q_{n-1}r_n$  and  $r_n | r_n$ .

If  $r_n$  divides  $r_{k+1}$  and  $r_k$ , then  $r_n$  divides  $r_{k-1} = q_{k+1}r_k + r_{k+1}$ ,

thus  $r_n$  divides  $r_k$  and  $r_{k-1}$ . Inductively,  $r_n$  divides  $b = r_{-1}$  and  $a = r_{-2}$ .

This proves that  $(a, b) \subseteq (r_n)$ . To show the reverse inclusion,

$a = q_0b + r_0$  implies  $r_0 \in (a, b)$  and  $b = q_1r_0 + r_1$  implies  $r_1 \in (b, r_0) \subseteq (a, b)$ .

And, if  $r_{k-1}, r_k \in (a, b)$ , then  $r_{k-1} = q_{k+1}r_k + r_{k+1}$  implies  $r_{k+1} \in (a, b)$ .

That is,  $r_k, r_{k+1} \in (a, b)$ . Inductively,  $r_n \in (a, b)$ .  $\square$

Def. An Integral Domain  $R$  is called Principal Ideal Domain if:  
Every Ideal in  $R$  is principal.

Directly, Euclidean domain  $\Rightarrow$  P.I.D.

Prop. Let  $R$  be a P.I.D., and let  $a$  and  $b$  be non-zero elements of  $R$ .

Let  $d \in R$  s.t.  $(d) = (a, b)$ . Then,

- 1)  $d$  is a g.c.d. of  $a$  and  $b$
- 2)  $d = ax + by$  for some  $x, y \in R$
- 3)  $d$  is unique up to multiplication by a unit of  $R$ .

Prop. Every nonzero prime ideal in a P.I.D is a Maximal.

Proof. Let  $(p) \subsetneq R$  be a Prime Ideal,  $p \neq 0$ . And let  $m \in R$  s.t.

$$(p) \subseteq (m) \subseteq R.$$

Enough to show that  $(m) = (p)$  or  $(m) = R$ . By the assumption,  $p = rm$  for some  $r \in R$ .

And, since  $(p)$  is prime,  $rm \in (p)$  implies  $r \in (p)$  or  $m \in (p)$ .

If  $r \in (p)$ , then  $r = ps$  for some  $s \in R$ ,  $p = rm = psm$ . Thus  $sm = 1$ ,  $m$  is unit,  $(m) = R$ .

If  $m \in (p)$ , then  $(m) \subseteq (p)$ , thus  $(m) = (p)$ . ~~pro~~

The Converse is true in general integral domain.

Def. Let  $R$  be an Integral domain, let  $r \in R$  be a nonzero, not a unit.

1). A nonzero, not unit  $r \in R$  is called irreducible in  $R$  if:

if  $r = ab$  with  $a, b \in R$ , then either  $a$  or  $b$  is a unit in  $R$ .

2). A nonzero  $p \in R$  is called prime in  $R$  if:  $(p) \subseteq R$  is prime ideal.

Equivalently,  $p \in R$  is prime  $\Leftrightarrow p$  is nonzero, not unit,  $p|ab \Rightarrow p|a$  or  $p|b$ .

Lemma. Let  $R$  be an integral domain, and  $r \in R$  be an element. Then,

$r$  is irreducible element if and only if there is no element  $m \in R$  s.t.  $(r) \subsetneq (m) \subsetneq R$ .

Proof,  $r$  is irreducible  $\Rightarrow$  If  $r = mx$ , then either  $m$  or  $x$  is unit

$\Rightarrow (m) = R$  or  $(m) = (r)$

$r$  is reducible  $\Rightarrow r = m \cdot n$  for some nonzero, nonunit  $m, n \in R \Rightarrow (r) \subsetneq (m) \subsetneq R$ .

Therefore,  $r \in R$  is irreducible  $\Leftrightarrow (r)$  is maximal principal ideal.  $\uparrow$   $n$  non unit  $\uparrow$   $m$  non unit

(But, it is not necessarily  $(r)$  is maximal ideal.)

Prop. Prime element is irreducible in an integral domain.

Proof. Let  $p \in R$  be a prime element in an integral domain.

If  $p = ab$ , then  $p = ab|ab$ , thus  $p|a$  or  $p|b$ . Either  $p|a$  or  $p|b$ ,

$\begin{cases} p = prb & \text{if } p|a, \\ p = asp & \text{if } p|b \end{cases}$

for some  $r, s \in R$ , which implies  $rb = 1$  or  $as = 1$ ,  $b$  or  $a$  is a unit.  $\square$

Prop. Irreducible element in P.I.D is a prime.

Proof. Let  $r \in R$  be an irreducible element in a P.I.D.

Suppose that an Ideal  $M \subseteq R$  satisfies  $(r) \subseteq M \subseteq R$ . Since  $R$  is P.I.D, let  $M = (m)$ .

Since  $r \in (m)$ ,  $r = mx$  for some  $x \in R$ . And,  $r$  is irreducible, thus either  $m$  or  $x$  is a unit.

If  $m$  is a unit, then  $(m) = R$ , if  $x$  is a unit, then  $m = rx^{-1} \in (r)$ . Thus  $(r) = (m)$ .

Therefore,  $(r)$  is Maximal, thus prime.

$\xleftarrow{\text{P.I.D.}}$   $\xleftarrow{\text{P.I.D.}}$   
 In Sum, in P.I.D., Maximal  $\Leftrightarrow$  Prime  $\Leftrightarrow$  Irreducible.  
 $\xrightarrow{\hspace{1cm}}$   $\xrightarrow{\hspace{1cm}}$

More observation for principal Ideal.

Let  $R$  be a Ring, Not Necessary identity, Commutative, no zero division.

Generally, if  $I, J \subseteq R$  are Ideal, then  $I+J = (I, J)$ , the smallest ideal containing  $I$  and  $J$ .

And, if  $I, J, K \subseteq R$  are Ideals in a Ring  $R$  with identity  $1$ ,

General Ring

$$I(J+K) = IJ + IK \text{ and } (I+J)K = IK + JK$$

Proof. Let  $x \in I(J+K)$ . Then  $x = a_1 b_1 + \dots + a_n b_n$ ,  $a_i \in I$ ,  $b_i \in J+K$

$$= a_1 (j_1 + k_1) + \dots + a_n (j_n + k_n), \quad j_i \in J, k_i \in K$$

$$= a_1 j_1 + \dots + a_n j_n + a_1 k_1 + \dots + a_n k_n \in IJ + JK.$$

And, since  $J, K \subseteq J+K$ ,  $IJ, IK \subseteq I(J+K)$ , thus  $IJ+IK \subseteq I(J+K)$ . ~~is~~ Ring with 1.  
Need identity?

This allows to us that: for any  $a, b, c \in R$  where  $R$  contains  $1$ ,

$$(a, b) = (a) + (b)$$

$$(a)[(b)+(c)] = (a)(b) + (a)(c) = (ab) + (ac)$$

$$[(a)+(b)](c) = (a)(c) + (b)(c) = (ac) + (bc)$$

Thm. Let  $R$  be a Ring, and  $I, J \subseteq R$  be Ideals. Then,

$$(I, J) \stackrel{\text{def}}{=} \bigcap_{\substack{S \text{ Ideal} \\ I, J \subseteq S}} S = I+J.$$

Proof. 1). The sum of ideals is Ideal.

Let  $a+b \in I+J$  be given, and let  $r \in R$ . Then,

$$r \cdot (a+b) = ra + rb \in I+J \text{ and } (a+b) \cdot r = ar + br \in I+J.$$

And, for  $a_1, a_2 \in I$  and  $b_1, b_2 \in J$ ,  $e_1, e_2 \in J$ , by Ideal

$$(a_1 + b_1) \cdot (a_2 + b_2) = \underbrace{a_1 a_2}_{\in I} + \underbrace{a_1 b_2 + b_1 a_2}_{\in J} + b_1 b_2 \in I+J.$$

by  $I, J$  are subring

And,  $I, J \subseteq I+J$  clearly, so  $(I, J) \subseteq I+J$ . And,  $a \in I, b \in J \Rightarrow a+b \in (I, J)$ .

Def. A Unique Factorization Domain is an integral domain  $R$  in which every nonzero, non-unit  $r \in R$  satisfies the following two properties:

1.  $r$  can be written as a finite product of irreducibles  $p_i \in R$ . (Not necessary distinct)
2. This representation is unique up to associates.

Prop. In a U.F.D., prime  $\Leftrightarrow$  irreducible.

Proof. Generally, prime element is irreducible. Let  $p \in R$  be an irreducible in a U.F.D.  $R$ .

Assume that  $p \mid ab$  for some  $a, b \in R$ . That is,  $ab = pc$  for some  $c \in R$ .

Since  $R$  is U.F.D. and  $a, b$  are nonzero, not unit,  $a$  and  $b$  have factorizations:  
(If either  $a$  or  $b$  is unit, then  $p$  also must divide other one)

$$a = u \cdot p_1 \cdots p_n, \quad b = v \cdot q_1 \cdots q_m \quad (u, v \text{ are units, } p_i, q_j \text{ are irreducible})$$

Now,  $ab = uv p_1 \cdots p_n q_1 \cdots q_m = pc$

and this factorization is unique up to associate,  $p$  must associate <sup>to</sup> one of  $p_1, \dots, p_n, q_1, \dots, q_m$ .

Thus,  $p \mid a$  or  $p \mid b$ .  ~~$p \mid c$~~

In sum, in U.F.D.,

|                                                               |                                                  |                                                             |                                                              |                                                                   |
|---------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------|
| $\begin{array}{c} \text{Maximal} \\ \updownarrow \end{array}$ | $\begin{array}{c} ? \\ \updownarrow \end{array}$ | $\begin{array}{c} \text{Prime} \\ \updownarrow \end{array}$ | $\begin{array}{c} \text{U.F.D.} \\ \updownarrow \end{array}$ | $\begin{array}{c} \text{Irreducible} \\ \updownarrow \end{array}$ |
| $\Leftrightarrow$                                             | $\Leftrightarrow$                                | $\Leftrightarrow$                                           | $\Leftrightarrow$                                            | $\Leftrightarrow$                                                 |

Thm. Every Principal Ideal Domain is Unique Factorization Domain.

Proof. Let  $R$  be a P.I.D., and  $r \in R$  be a nonzero, not unit.

If  $r$  is irreducible, then clear. Suppose  $r$  is reducible. Then,

$r = r_1 r_2$  for some nonzero  $r_1, r_2$ . If both  $r_1$  and  $r_2$  are irreducible, clear.

If either  $r_1$  or  $r_2$  is not irreducible, wlog,  $r_1$  is reducible,

$r_1 = r_{11} r_{12}$  for some nonzero  $r_{11}, r_{12}$ . Clearly,

$$(r) \subsetneq (r_1) \subsetneq (r_{11}) \subsetneq R$$

because  $r \in (r_1)$  since  $r = r_1 r_2$ , but  $r_1 \notin (r)$ , since  $r_2$  is not unit.

(Specifically, if  $r_1 \in (r)$ , then  $r_1 = r x = r_1 r_2 x \Rightarrow r_1 (1 - r_2 x) = 0 \Rightarrow r_2 x = 1 \Rightarrow r_2$  unit.)

Inductively, we can construct an ascending chain of ideals:

$$(r) \subsetneq (r_1) \subsetneq (r_{11}) \subsetneq \dots \subsetneq R$$

But, Every Ascending Chain in P.I.D will be stationary. Because:

Let  $\{I_n\}_{n=1}^{\infty}$  be a collection of Ideals of P.I.D  $R$  s.t

$$I_1 \subseteq I_2 \subseteq \dots \subseteq R.$$

Denote  $I = \bigcup_{n=1}^{\infty} I_n$ . Then, it is Ideal, because for any  $r \in R$  and  $a \in I$ ,  $a \in I_n$  for some  $n \in \mathbb{N}$ .

Thus  $ra \in I_n \subseteq I$ . Also,  $I$  is a subring, because  $\{I_n\}$  is nested. Thus  $I$  is ideal.

Since  $R$  is P.I.D, denote  $I = (d)$  for some  $d \in R$ . Since  $d \in I_1$ ,  $d \in I_n$  for some  $n \in \mathbb{N}$ .

This means  $I_n \subseteq I = (d) \subseteq I_n$ , thus  $I_n = I$ .  $\perp$  P.I.D  $\Rightarrow$  Noetherian Domain.

Therefore there is No infinite strictly ascending chain,

thus  $r$  has a finite irreducible factorization. Now the Uniqueness remains.

Using induction for number of irreducible factors for some factorization of  $r$ .

If  $n=1$ , then  $r$  is irreducible. If  $r = p_1 p_2 \dots p_m$  is other factorization where  $p_i$  are irreducible, then  $r$  must divide one of  $p_i$ , because  $r$  is prime. That is,  $p_i = u \cdot r$  for some  $i$ .

Wlog, put  $i=1$ . Then,  $r = u r p_2 \dots p_m \Rightarrow r(1 - u p_2 \dots p_m) = 0$ , but since  $p_2, \dots, p_m$  are not units,

it can not be possible. Suppose for  $n-1$ , the Uniqueness holds. If  $r$  has two factorization

$$r = p_1 p_2 \dots p_n = q_1 q_2 \dots q_m \quad m \geq n$$

Similarly,  $p_1$  must divide one of  $q_i$ , wlog  $q_1$ . Then  $p_2 \dots p_n = u q_2 \dots q_m$ , for some unit  $u$ .

These are same representation, by the induction.

Def. An integral domain  $R$  in which every ideal generated by two elements is principal is called a Bézout Domain. Simply, for all  $a, b \in R$ , there exists  $d \in R$  s.t.

$$(a, b) = (a) + (b) = (d)$$

Prop. Let  $R$  be an integral domain.

$R$  is Bézout Domain if and only if for any  $a, b \in R$ ,  $d = \gcd(a, b)$  exists with  $d = ax + by$  for some  $x, y \in R$ .

Proof. Suppose  $R$  is Bézout domain. Let  $a, b \in R$  be given. Then,

$$(a, b) = (d) \text{ for some } d \in R.$$

Now,  $d = ax + by$  for some  $x, y \in R$ , and, since  $(a), (b) \subseteq (a, b) = (d)$ ,  $d \mid a$  and  $d \mid b$ . And, if  $d' \mid a$  and  $d' \mid b$ , then  $d' \mid d (= ax + by)$ . The converse is trivial.

Thm. Every finitely generated Ideal of a Bézout domain is principal.

Proof. Let  $R$  be a Bézout domain, and  $I \subseteq R$  be a finitely generated Ideal. Denote

$$I = (a_1, \dots, a_n) \text{ for some } a_1, \dots, a_n \in R.$$

Using induction for  $n$ . If  $n = 1$  or  $2$ , trivial. Suppose for  $n-1$ , the statement holds.

Then,  $(a_1, \dots, a_{n-1}) = (p)$  for some  $p \in R$ . Enough to show  $(a_1, \dots, a_n) = (p, a_n)$ .

Let  $a_1x_1 + \dots + a_nx_n \in (a_1, \dots, a_n)$  be given. Since for each  $a_i$ ,  $p \mid a_i$ , thus

$$a_1x_1 + \dots + a_{n-1}x_{n-1} + a_nx_n = px_1 + a_nx_n \in (p, a_n).$$

Converse is trivial. ~~or~~ Simply,  $(p, a_n) = (p) + (a_n) = (a_1, \dots, a_{n-1}) + (a_n) = (a_1, \dots, a_n)$ .

Thm. If  $R$  is Bézout domain and also U.F.D., then it is P.I.D.

Proof. Let  $I \subseteq R$  be an Ideal of  $R$ , and  $a \in I$  be a nonzero, minimal number of irr. factors.

If  $I \neq (a)$ , then there is  $b \in I \setminus (a)$ . Since  $R$  is Bézout domain, for some  $d \in I$ ,

$$(d) = (a, b) \subseteq I.$$

That is,  $d \mid a$  and  $d \mid b$ . But, since the minimality of  $a$ ,  $d$  must associates to  $a$ .

But, this implies  $d \mid b \Rightarrow b \in (d) = (a)$ , Contradiction. ~~or~~

Prop. In Bézout domain, every irreducible element generates Maximal Ideal, thus prime.

Proof. Let  $r \in R$  be an irreducible. Let  $M$  be an Ideal in  $R$  s.t.  $(r) \subseteq M \subseteq R$ .

Take  $x \in M \setminus (r)$ . [If no such  $x$  exists, then  $M = (r)$ , thus  $(r)$  is maximal  $\Rightarrow$  prime.]  
Clearly,  $(r) \neq (r, x) \subseteq M$ . Since  $R$  is a Bézout domain, there exists  $d \in R$  s.t.

$$(r, x) = (r) + (x) = (d)$$

But, since  $r$  is irreducible,  $d$  must be a unit. That is,  $(r, x) = R$

Consequently,  $(r)$  is Maximal, thus prime ideal.

+ More precisely,  $d|r \Leftrightarrow r = ds$  for some  $s \in R$ . Since  $r$  is irreducible, either  $d$  or  $s$  is a unit. If  $s$  is a unit, then

$$d = rs^{-1} | x \Rightarrow x = rs^{-1}t \in (r) \text{ for some } t \in R$$

But, it is Contradiction. Therefore  $d$  must be a unit.

Def. An Integral Domain  $R$  is called a Noetherian Domain if:

If a sequence of Ideals  $\{I_n\}$  in  $R$  satisfies that:

there are no infinite increasing chains of Ideals.

Thm. Let  $R$  be an Integral domain.

$R$  is Noetherian domain if and only if Every Ideal of  $R$  is finitely generated.

Proof. Suppose that  $R$  is Noetherian domain. Assume  $R$  has an ideal  $I$ , which cannot be generated finitely. Then, we can construct  $\{I_n\}$  as follows:

given  $a_1 \in I$ ,  $I_1 = (a_1) \subseteq I$

given  $a_2 \in I \setminus I_1$ ,  $I_2 = (a_1, a_2) \subseteq I$

given  $a_3 \in I \setminus I_2$ ,  $I_3 = (a_1, a_2, a_3) \subseteq I$

Then,  $I_1 \subsetneq I_2 \subsetneq \dots \subsetneq I$ , because  $a_n \notin I_{n-1}$ , so  $I_{n-1} \neq I_n$

But, this is Contradiction to  $R$  satisfies Acc on Ideals,

Conversely, let  $\{I_n\}_{n=1}^{\infty}$  be a chain of ideals in  $R$  s.t

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots$$

Take  $I = \bigcup_{n=1}^{\infty} I_n$ . Then, for any  $a, b \in I$ ,  $a, b \in I_N$  for some  $N \in \mathbb{N}$ .

Since  $I_N$  is a subring,  $a-b \in I_N \subseteq I$  and  $ab \in I_N \subseteq I$ , so  $I$  is subring, moreover, for any  $r \in R$ ,  $ra \in I_N \subseteq I$ , so it is an Ideal.

By the hypothesis,  $I$  is generated finitely. That is,

$$I = (a_1, \dots, a_n) \text{ for some } a_1, \dots, a_n \in I.$$

Since each  $a_i \in I$ , there is  $N_i \in \mathbb{N}$  s.t  $a_i \in I_{N_i}$ . Now, take  $m = \max \{N_1, \dots, N_n\}$ ,

$$I = (a_1, \dots, a_n) \subseteq I_m \subseteq I$$

Note: For a field  $F$ ,  $F[x_1, x_2, \dots]$  is Not Noetherian domain:

$$(x_1) \subsetneq (x_1, x_2) \subsetneq (x_1, x_2, x_3) \subsetneq \dots$$

But,  $F[x_1, x_2]$  is Noetherian, by the Hilbert basis Theorem.

Since  $F[x_1, x_2]$  is Not P.I.D., it is also Not Bezout.

$\mathbb{F}[x, y]$ .

Let  $F$  be a field, and let  $x, y$  be independent indeterminates.

1.  $F[x, y]$  is a Noetherian Domain.

Proof. Since  $F[x, y] = (F[x])[y]$ , the Hilbert basis Theorem gives that the result

2.  $F[x, y]$  is Not P.I.D.

Proof. Consider the ideal  $(x, y)$ . If  $(x, y)$  is principal then

$$(x, y) = (a(x, y)) \text{ for some } a(x, y) \in F[x, y]$$

There is,  $x = f(x, y) \cdot a(x, y)$ , for some  $f(x, y) \in F[x, y]$ .

If  $a(x, y)$  is constant, then  $(x, y)$  contains a unit, so it is entire. Contradiction.

So,  $a(x, y)$  must be  $a \cdot x$ , where  $a$  is non-zero constant.

Similarly,  $a(x, y) = b \cdot y$  for some non-zero  $b \in F$ ,  $ax = by$ , Contradiction.

3.  $F[x, y]$  is Not Bezout domain.

4.  $x$  and  $y$  in  $F[x, y]$  are relatively prime, but  $F[x, y] \neq (x) + (y)$ .

Example. Let  $R = \mathbb{Z}[x]$ . Then,  $(2, x) \subset R$  has the following properties:

$(2, x) = \{2p(x) + xq(x) \mid p, q \in \mathbb{Z}[x]\} \subsetneq R$ , since  $1 \notin (2, x)$ . Thus proper.

If  $(2, x) = (a(x))$  for some  $a(x) \in \mathbb{Z}[x]$ , then,  $2 \in (a(x))$ , this means

$$2 = a(x) \cdot p(x) \text{ for some } p(x) \in \mathbb{Z}[x].$$

Now,  $a(x), p(x) \in \{\pm 1, \pm 2\}$ . If  $a(x) = \pm 1$ , then  $(a(x)) = (1) = R$ , Contradiction. Thus  $a(x) = \pm 2$ . But,  $x \notin (\pm 2)$ , Contradiction. Now,  $(2, x)$  is not principal.

But  $(2, x) \subsetneq \mathbb{Z}[x]$  is Maximal, because a map

$$\varphi: \mathbb{Z}[x] \rightarrow \mathbb{Z}_2: p(x) \mapsto p(0) \bmod 2 \quad \mathbb{Z}[x] \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}_2.$$

is an onto Ring-Homomorphism having  $\ker \varphi = (2, x)$  because

$$p(x) \in \ker \varphi \Leftrightarrow p(0) \bmod 2 = 0 \Leftrightarrow p(x) \in (2, x)$$

Thus  $\mathbb{Z}[x]/\ker \varphi \cong \mathbb{Z}_2$  is a field,  $\ker \varphi = (2, x)$  a Maximal, further, prime. And, clearly,  $2 \in R$  is irreducible but  $(2) \subsetneq (2, x)$ , it is prime, but not Maximal.

$$(x) = \{p(x) \in \mathbb{Z}[x] \mid p(0) = 0\}.$$

Note:  $(2, x) \subset \mathbb{Z}[x]$  is also maximal, because:

$$\mathbb{Z}[x]/(2) = \mathbb{Z}[x]/2\mathbb{Z}[x] \cong (\mathbb{Z}/2\mathbb{Z})[x] \text{ with the natural Homomorphism}$$

$$\pi: \mathbb{Z}[x] \rightarrow (\mathbb{Z}/2\mathbb{Z})[x]: \sum a_n x^n \mapsto \sum (a_n \bmod 2) \cdot x^n$$

And consider the evaluation Homomorphism

$$\varphi: (\mathbb{Z}/2\mathbb{Z})[x] \rightarrow \mathbb{Z}/2\mathbb{Z}: p(x) \mapsto p(0)$$

Then,  $\ker \pi = 2\mathbb{Z}[x] = (2)$  and  $\ker \varphi = \{p(x) \in (\mathbb{Z}/2\mathbb{Z})[x] \mid p(0) = 0\} = (x)$

Moreover, since  $\pi$  and  $\varphi$  are surjective, so  $\varphi \circ \pi$  is also surjective,

$$\ker(\varphi \circ \pi) = \{p(x) \in \mathbb{Z}[x] \mid p(0) \bmod 2 = 0\} = (2, x)$$

$$p(x) = p(0) + x \cdot q(x) \text{ where } q(x) = (p(x) - p(0))/x \text{ or } 0.$$

$$\text{So, } p(0) \equiv 0 \bmod 2$$

$$\Rightarrow p(x) = 2k + x \cdot q(x) \in (2) + (x)$$

Example. The polynomial ring  $\mathbb{Z}[x]$  is Not P.I.D., but U.F.D.

Specifically,  $(2, x) \subset \mathbb{Z}[x]$  is Not Principal, but Maximal.

And,  $(2) \subset \mathbb{Z}[x]$  is Irreducible and prime, but Not Maximal.

Example. The quadratic integer ring  $\mathbb{Z}[\sqrt{-5}]$  is Not U.F.D., but Noether Domain.

Not U.F.D:  $6 = 2 \cdot 3 = (1 + \sqrt{-5}) \cdot (1 - \sqrt{-5})$  and

$2, 3, 1 \pm \sqrt{-5}$  are irreducible

Quadratic Integer Ring with Norm.

Def. Let  $D$  be a square free integer. Define

$$\mathbb{Z}[\sqrt{D}] \stackrel{\text{def}}{=} \{a + b\sqrt{D} \mid a, b \in \mathbb{Z}\}$$

is called a Quadratic Integer Ring. And, define a Norm on  $\mathbb{C}$  as

$$N: \mathbb{C} \rightarrow \mathbb{R}: a + ib \mapsto a^2 + b^2$$

or simply,  $N(\alpha) = \alpha \cdot \bar{\alpha}$ .

Prop.  $N(\alpha) = 0$  if and only if  $\alpha = 0$

Proof.  $N(\alpha) = 0 \Leftrightarrow a^2 + b^2 = 0$  where  $\alpha = a + ib$ . This is clear.

Prop.  $N(\alpha\beta) = N(\alpha)N(\beta)$

Proof. Denote  $\alpha = a + ib$ ,  $\beta = c + id$ .

$$\begin{aligned} N(\alpha\beta) &= N(ac - bd + i(ad + bc)) \\ &= (ac - bd)^2 + (ad + bc)^2 \end{aligned}$$

$$= a^2c^2 + b^2d^2 - 2abcd + a^2d^2 + b^2c^2 + 2abcd$$

$$= (a^2 + b^2)(c^2 + d^2) = N(\alpha)N(\beta)$$

Example. The Quadratic Integer  $\mathbb{Z}[\sqrt{-5}]$ .

1).  $\mathbb{Z}[\sqrt{-5}]$  has no zero divisor. (Thus Integral domain)

Proof. If  $\alpha, \beta \in \mathbb{Z}[\sqrt{-5}]$  satisfies  $\alpha\beta = 0$ , then

$$N(\alpha\beta) = N(\alpha)N(\beta) = 0$$

This implies  $N(\alpha) = 0$  or  $N(\beta) = 0$ , that is,  $\alpha = 0$  or  $\beta = 0$ .

2).  $\mathbb{Z}[\sqrt{-5}]$  has units only  $\pm 1$ .

Proof. Observe that  $\alpha\beta = 1 \Rightarrow N(\alpha\beta) = N(\alpha)N(\beta) = 1$ .

Since  $N[\mathbb{Z}[\sqrt{-5}]] \subseteq N(\mathbb{U}\{0\})$ ,  $N(\alpha) = N(\beta) = 1$

But,  $N(\alpha) = a^2 + 5b^2 = 1$  iff  $a = \pm 1$  and  $b = 0$ .

Thus  $\alpha = \beta = 1$  or  $\alpha = \beta = -1$ .

3).  $2, 3, 1 \pm \sqrt{-5}$  are irreducible, and no two which are not associate.

Proof. If 2 is reducible, then  $2 = \alpha\beta$  for some  $\alpha, \beta \in \mathbb{Z}[\sqrt{-5}]$  s.t.  $\alpha, \beta \neq \pm 1$ .

But,  $4 = N(2) = N(\alpha\beta) = N(\alpha)N(\beta)$  implies  $N(\alpha) = 1, N(\beta) = 4$  or  $N(\alpha) = N(\beta) = 2$  or  $N(\alpha) = 4, N(\beta) = 1$

Since  $N(\alpha) = 1$  iff  $\alpha = \pm 1$ , the only possible case is  $N(\alpha) = N(\beta) = 2$ . But,

$a^2 + 5b^2 = 2$  has no integer solution. Similarly, 3 is irreducible.

(But, 5 is reducible, since  $5 = (\sqrt{-5}) \cdot (-\sqrt{-5})$ )

And, if  $1 \pm \sqrt{-5}$  is reducible, then  $1 \pm \sqrt{-5} = \alpha\beta$  with  $\alpha, \beta \neq \pm 1$ . Then,

$$6 = N(1 \pm \sqrt{-5}) = N(\alpha\beta) = N(\alpha)N(\beta)$$

Then, since  $N(\alpha), N(\beta) \neq 1$ ,  $N(\alpha) = 2, N(\beta) = 3$  or  $N(\alpha) = 3, N(\beta) = 2$ .

But,  $a^2 + 5b^2 = 2$  or 3 have no integer solution, thus  $1 \pm \sqrt{-5}$  are irreducible.

And, not associated is clear, because  $\mathbb{Z}[\sqrt{-5}]$  has unit only  $\pm 1$ .

4).  $\mathbb{Z}[\sqrt{-5}]$  is Not U.F.D.

$$6 = 2 \cdot 3 = (1 + \sqrt{-5}) \cdot (1 - \sqrt{-5}).$$

5).  $2 \in \mathbb{Z}[\sqrt{-5}]$  is irreducible but not prime.

$$2 \mid 6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5}), \text{ but } 2 \nmid 1 \pm \sqrt{-5}.$$

5).  $\mathbb{Z}[\sqrt{-5}]$  is Noether domain. (Thus, not Bézout)

$$\mathbb{Z}[x, y]$$

Define a homomorphism:

$$\varphi: \mathbb{Z}[x] \rightarrow \mathbb{Z}[\sqrt{-5}]; p(x) \mapsto p(\sqrt{-5})$$

$$\underbrace{\mathbb{Z}[x]}_{\text{U.F.B.}} / \underbrace{(x^2+5)}_{\substack{\text{irr.} \\ \downarrow \\ \text{Prime}}} \cong \mathbb{Z}[\sqrt{-5}]$$

$$\mathbb{Z} \text{ E.D.} \Rightarrow \text{P.I.D.} \Rightarrow \text{Noether} \Rightarrow \mathbb{Z}[x] \text{ noether} \Rightarrow \mathbb{Z}[\sqrt{-5}] \text{ noether}$$

6).  $11 \in \mathbb{Z}[\sqrt{-5}]$  is prime element, but  $(11)$  is not Maximal.  
Let  $a+b\sqrt{-5} \in \mathbb{Z}[\sqrt{-5}]$

7).  $1+\sqrt{-5}$  is not prime,

$$\text{observe that } (1+\sqrt{-5})(1-\sqrt{-5}) = 1-(-5) = 6 = 2 \cdot 3,$$

If  $1+\sqrt{-5}$  divides 2, then  $N(2) = 4 = N(1+\sqrt{-5}) \cdot N(\alpha) = 6 \cdot N(\alpha)$ , there is no such  $\alpha$ .

If  $1+\sqrt{-5}$  divides 3, then  $9 = 6 \cdot N(\alpha)$ , similarly, there is no such  $\alpha$ .

Example. The Quadratic Integer Ring  $\mathbb{Z}[\sqrt{-5}]$

1).  $\mathbb{Z}[\sqrt{-5}]$  has no zero divisor.

Suppose that  $(a+b\sqrt{-5})(c+d\sqrt{-5}) = ac - 5bd + (ad+bc)\sqrt{-5} = 0$ .

It equals to:  $\begin{cases} ac - 5bd = 0 \\ ad + bc = 0 \end{cases}$ . Then,

$$ad = 5bd^2 \Leftrightarrow -bdc^2 = 5bd^2 \Leftrightarrow bd(5d - c^2) = 0.$$

Thus,  $b=0$  or  $d=0$  or  $5d = c^2$ .

$$b=0 \Rightarrow ac=0 \Rightarrow \begin{cases} a=0 \Rightarrow a+b\sqrt{-5}=0 \\ c=0 \Rightarrow ad=0 \Rightarrow \begin{cases} a=0 \Rightarrow a+b\sqrt{-5}=0 \\ d=0 \Rightarrow c+d\sqrt{-5}=0 \end{cases} \end{cases}$$

$$d=0 \Rightarrow ac=0 \Rightarrow \begin{cases} a=0 \Rightarrow bc=0 \Rightarrow \begin{cases} b=0 \Rightarrow a+b\sqrt{-5}=0 \\ c=0 \Rightarrow c+d\sqrt{-5}=0 \end{cases} \\ c=0 \Rightarrow c+d\sqrt{-5}=0 \end{cases}$$

$$5d = c^2 \Rightarrow ac - 5bd = ac - dc^2 = c(a - dc) = 0 \\ \Rightarrow \begin{cases} c=0 \Rightarrow c+d\sqrt{-5}=0 \\ a=dc \Rightarrow adc + bc^2 = a^2 + 5bd = 0 \end{cases}$$

1). Only unit of  $\mathbb{Z}[\sqrt{-5}]$  is  $\pm 1$ .

Suppose that  $(a+b\sqrt{-5})(c+d\sqrt{-5}) = ac - 5bd + (ad+bc)\sqrt{-5} = 1$ .

It equals to:  $\begin{cases} ac - 5bd = 1 \\ ad + bc = 0 \end{cases}$ .

Fail Try

Fail try

And, let  $p \notin R$  be a prime ideal. Let  $M$  be an Ideal s.t.  $p \subseteq M \subseteq R$ .

$$\begin{aligned}(a, b) &\subseteq (a), (b) \\ &\subseteq (a) + (b)\end{aligned}$$

$$(a, b) = (a) + (b) = (d)$$

To show  $r$  is prime, we will show that  $(r)$  is maximal.

Suppose that  $r \mid ab$  for some  $a, b \in R$ .

That is,  $ab = rx$  for some  $x \in R$ .

Since  $R$  is Bézout domain, there exists  $d \in R$  s.t.  $(d) = (a, b)$

Suppose that  $r \nmid ab$ . That is,  $ab \notin (r)$ .

Since  $R$  is Bézout domain,  $(a) + (b) = (d)$  for some  $d \in R$ , and  $(a, b) \subseteq (a), (b)$  thus

$$(a, b) \subseteq (a) + (b) = (d)$$

But, since  $d$  is greatest common divisor of  $a$  and  $b$ ,

$$(r) \not\subseteq (d)$$

$r \nmid a$  and  $r \nmid b$

$$d = r$$

$$d \mid ab$$